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## Session 2 — Linear algebra II

### Exercise sheet

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*Determinant, rank, eigenvalues, symmetric matrices, PCA and the SVD. The recurring picture is the same data table as last time — rows are participants, columns are measures — and the question of the day is: how many directions does it really have? Show your reasoning; most answers are short.*

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#### Exercise 1 Determinants and inverses.

- (a) Compute  $\det \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$  and the inverse.
- (b) Compute  $\det \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$  and the inverse.
- (c) Compute  $\det \begin{pmatrix} 4 & 6 \\ 2 & 3 \end{pmatrix}$ . Is the matrix invertible? What is its rank?
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#### Exercise 2

- (a) For which  $k$  is  $\begin{pmatrix} 1 & 2 \\ 2 & k \end{pmatrix}$  non-invertible, and what happens to its rank there?
- (b) For which  $k$  is  $\begin{pmatrix} 1 & 2 \\ 3 & k \end{pmatrix}$  non-invertible?
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#### Exercise 3 The determinant as a signed area factor.

- (a)  $\det \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -1$ . What do the size and the sign each mean geometrically?
- (b) With  $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$  and  $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , check that  $\det(AB) = \det A \det B$ .
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#### Exercise 4 Rank.

- (a) What is the rank of  $\begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}$ ? Does it have an inverse, and on what line do all its outputs lie?
- (b) What is the rank of  $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$ ?

- (c) Give a  $2 \times 2$  matrix of rank 0, one of rank 1, and one of rank 2.

**Exercise 5** *A table has a redundant column whenever one column is a linear combination of the others. The rank is then smaller than the number of columns, and no analysis can recover a unique weight for each column.*

A study records, for each participant, the score on item 1, the score on item 2, and the total of the two:

$$X = \begin{pmatrix} 4 & 9 & 13 \\ 6 & 7 & 13 \\ 5 & 8 & 13 \\ 9 & 4 & 13 \\ 7 & 6 & 13 \end{pmatrix}.$$

- (a) Give the rank of  $X$ . Justify in one line.
- (b) The weights  $\mathbf{w} = (1, 1, 0)$  and  $\mathbf{w}' = (0, 0, 1)$  are different. Show that  $X\mathbf{w} = X\mathbf{w}'$ .
- (c) In one sentence: what does that mean for a regression using all three columns as predictors?

**Exercise 6** *An outer product  $\mathbf{m}\mathbf{n}^\top$  is the simplest non-trivial matrix there is. Read as a table it says: everybody has the same profile  $\mathbf{n}$ , only the overall level  $m_i$  differs.*

Compute the outer product  $\begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 3 & -1 \end{pmatrix}$ , then its determinant and its rank.

### Exercise 7

- (a) A  $200 \times 20$  table of responses is exactly  $X = \mathbf{m}\mathbf{n}^\top$ , with  $\mathbf{m} \in \mathbb{R}^{200}$  and  $\mathbf{n} \in \mathbb{R}^{20}$ . What is its rank? How many numbers would you need to store instead of the 4000 entries?
- (b) Now  $X = \mathbf{m}_1\mathbf{n}_1^\top + \mathbf{m}_2\mathbf{n}_2^\top$ , with  $\mathbf{m}_1, \mathbf{m}_2$  independent and  $\mathbf{n}_1, \mathbf{n}_2$  independent. Give the rank, and describe in words what the table looks like.

### Exercise 8

Eigenvectors by inspection.

- (a) Verify that  $(1, 0)$  is an eigenvector of  $\begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$  and give its eigenvalue.
- (b) Give the eigenvalues of  $\begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}$  and of  $\begin{pmatrix} 5 & 0 \\ 0 & -2 \end{pmatrix}$ , with their eigenvectors.

### Exercise 9

Use the trace  $T$  and determinant  $D$ , and the characteristic equation  $\lambda^2 - T\lambda + D = 0$ .

- (a)  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ : give  $T$ ,  $D$ , the eigenvalues and the eigenvectors.

- (b)  $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ : give  $T$ ,  $D$  and the eigenvalues.
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**Exercise 10** Find the eigenvalues and eigenvectors of:

- (a)  $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ ;  
(b)  $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ ;  
(c)  $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ ;  
(d)  $\begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}$ .
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**Exercise 11** Complex eigenvalues.

- (a) Find the eigenvalues of  $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$  and of  $\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$ . Real or complex, and what do they mean geometrically?  
(b) For which real  $a$  does  $\begin{pmatrix} 1 & a \\ -a & 1 \end{pmatrix}$  have complex eigenvalues?
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**Exercise 12** Show that  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  is *not* diagonalizable. (*Find its eigenvalues first, then count independent eigenvectors.*)

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**Exercise 13** *Applying the same linear rule over and over — a learning rule trial after trial, an iterative algorithm — is governed entirely by the eigenvalues.*

- (a) A matrix has eigenvalues 2 and 0.5. Apply it many times to a generic vector: which eigendirection does the result line up with, and why?  
(b) With  $A = P\Lambda P^{-1}$  and  $\lambda_1 = 1$ ,  $\lambda_2 = \frac{1}{2}$ , describe  $A^{10}$ : which mode survives?  
(c) Using the eigendecomposition found in Exercise 10(b), compute  $A^3$  for  $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ , and check it against direct multiplication.
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**Exercise 14** A  $2 \times 2$  matrix has eigenvalues  $-3$  and  $5$ . Give its trace and its determinant,

without knowing the matrix.

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**Exercise 15** Symmetric matrices.

- (a) Check that  $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$  is symmetric, find its eigenvalues, and verify that its two eigenvectors are orthogonal.
- (b) Do the same for  $\begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}$ .
- (c) True or false: a real symmetric matrix can have complex eigenvalues. Justify.
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**Exercise 16** *For centred data,  $C = \frac{1}{n}X^T X$  is the covariance matrix: variances on the diagonal, co-variation off it. It is symmetric and positive semi-definite, its eigenvectors are the principal components, and the matching eigenvalues are the variances along them.*

- (a) A covariance matrix has eigenvalues 5 and 0.2. Along which eigenvector does the data mostly spread, and what fraction of the total variance does that direction carry?
- (b) For  $C = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ , give the two principal components in words (the two measures are the scores on two similar tasks), and the fraction of variance along the first. Check that  $\text{tr } C$  is the sum of the eigenvalues.
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**Exercise 17** Which of these could be a covariance matrix? Justify each answer with the eigenvalues.

- (a)  $\begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$  — and which axis is then the first principal component?
- (b)  $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$  — and what is its rank?
- (c)  $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ ;
- (d)  $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ .
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**Exercise 18** Four participants, two measures, already centred:  $(1, 1)$ ,  $(-1, -1)$ ,  $(2, 2)$ ,  $(-2, -2)$ .

- (a) Compute  $C = \frac{1}{4} \sum_i \mathbf{x}_i \mathbf{x}_i^T$ .
- (b) Give its eigenvalues and the first principal component.

(c) What does the second eigenvalue tell you about the two measures?

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**Exercise 19** Now the four centred points are  $(2, 0)$ ,  $(0, 2)$ ,  $(-2, 0)$ ,  $(0, -2)$ . Compute  $C$  and say what is special about its principal components.

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**Exercise 20** A data table is  $n \times p$  and almost never square, so it has no eigenvalues at all. The singular value decomposition  $X = U\Sigma V^T$  exists for every matrix:  $U$  and  $V$  are orthogonal,  $\Sigma$  is diagonal with  $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ .

- (a) A centred table  $X$  is  $200 \times 6$ . Give the dimensions of  $U$ ,  $\Sigma$  and  $V$ , and say how many singular values there are.
  - (b) Which of the three factors holds the directions in *measure* space (i.e. the principal components)? Which holds one row per participant?
  - (c) Why can we not simply take eigenvalues of  $X$  instead?
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**Exercise 21** Starting from  $X = U\Sigma V^T$  with  $U^T U = I$  and  $V^T V = I$ :

- (a) Show that  $X^T X = V\Sigma^2 V^T$ .
  - (b) Deduce the relationship between the singular values  $\sigma_k$  of  $X$  and the eigenvalues  $\lambda_k$  of the covariance matrix  $C = \frac{1}{n} X^T X$ .
  - (c) Show similarly that  $XX^T = U\Sigma^2 U^T$  — so  $U$  collects the eigenvectors of the *participant*  $\times$  *participant* matrix.
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**Exercise 22** Four participants, two measures, already centred:

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \\ -1 & -1 \end{pmatrix}.$$

- (a) Compute  $X^T X$ , then its eigenvalues, then the singular values of  $X$ .
  - (b) Give  $\mathbf{v}_1$  and compute  $\mathbf{u}_1 = X\mathbf{v}_1/\sigma_1$ . Check  $\|\mathbf{u}_1\| = 1$ .
  - (c) Verify that  $X = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T$  exactly, and give the rank of  $X$ .
  - (d) What fraction of the variance does the first component carry? Interpret the answer in one sentence.
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**Exercise 23** A centred table  $X$  has singular values  $\sigma = (6, 3, 2, 1)$ .

- (a) What fraction of the variance is carried by the first component? By the first two?

- (b) Write down the best rank-2 approximation  $X_2$  in terms of the  $\mathbf{u}_k$ ,  $\mathbf{v}_k$  and  $\sigma_k$ , and give the rank of the error  $X - X_2$ .
  - (c) A colleague reports “the first two components explain 75% of the variance” using  $\sigma$  rather than  $\sigma^2$ . Recompute their number and say why it is wrong.
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**Exercise 24** Twenty questionnaire items are answered by 500 people. The singular values of the centred table drop sharply after the second one.

- (a) Roughly what rank does the table have, and roughly what rank does it *effectively* have? Are these the same question?
- (b) How many numbers does the rank-2 truncation need, compared with the 10 000 entries of the table?
- (c) In one sentence, say what the truncation is *not* entitled to claim.